

CYBERSECURITY ANALYST

Complete Career Roadmap

From Zero to Senior Analyst · 2025 Edition

[SOC Analyst] [Threat Intelligence] [Incident Response] [Penetration Testing] [Cloud Security] [Compliance]

4–6	15+	\$85K–\$150K	Top 10
Years to Senior	Certifications	Avg. Salary	Fastest Growing

ROADMAP OVERVIEW

A cybersecurity analyst protects organizations from digital threats by monitoring networks, investigating incidents, and hardening defenses. This roadmap provides a structured, phase-by-phase guide — from foundational IT knowledge through advanced threat hunting and leadership roles.

CORE COMPETENCY IMPORTANCE



LEARNING PHASES

Phase 1 IT & Networking Foundations 3–6 Months

- TCP/IP & OSI Model Understand how data flows through networks
- Linux Fundamentals CLI, file permissions, processes, logs
- Windows Administration Active Directory, Group Policy, Event Viewer
- Basic Scripting Python or Bash for automation basics
- Virtualization VMware / VirtualBox for lab environments

Recommended Certs: ■ CompTIA A+ ■ Google IT Support

Phase 2 Core Security Concepts 4–6 Months

- CIA Triad & Frameworks NIST, ISO 27001, CIS Controls overview
- Cryptography Basics Symmetric, asymmetric, PKI, TLS/SSL
- Vulnerability Management CVE, CVSS scoring, patch management
- Firewalls & IDS/IPS pfSense, Snort, Suricata configuration
- Network Traffic Analysis Wireshark, tcpdump packet inspection

Recommended Certs: ■ CompTIA Security+ ■ CompTIA Network+

Phase 3 Security Operations Center (SOC) 6–9 Months

- SIEM Platforms Splunk, Microsoft Sentinel, IBM QRadar
- Log Analysis Windows/Linux event logs, syslog, CEF
- Alert Triage False positive reduction, playbooks
- Threat Intelligence IOCs, TTPs, MITRE ATT&CK; framework
- Ticketing & Reporting ServiceNow, JIRA, incident documentation

Recommended Certs: ■ CompTIA CySA+ ■ Splunk Core User ■ Blue Team Labs Online

Phase 4	Incident Response & Forensics	4–6 Months
■ IR Lifecycle	Preparation, detection, containment, recovery	
■ Digital Forensics	Disk imaging, memory analysis, FTK/Autopsy	
■ Malware Analysis	Static & dynamic analysis, sandbox tools	
■ Threat Hunting	Hypothesis-driven proactive hunting	
■ Evidence Handling	Chain of custody, legal considerations	
Recommended Certs: ■ GCIH ■ GCFE ■ EC-Council CHFI		

Phase 5	Offensive Security & Red Team	4–6 Months
■ Penetration Testing	Methodology, scoping, rules of engagement	
■ Web App Security	OWASP Top 10, Burp Suite, SQLi, XSS	
■ Active Directory Attacks	Kerberoasting, pass-the-hash, BloodHound	
■ CTF & Lab Practice	HackTheBox, TryHackMe, PicoCTF	
■ Report Writing	Executive & technical findings, remediation	
Recommended Certs: ■ eJPT ■ CompTIA PenTest+ ■ OSCP		

Phase 6	Cloud & Advanced Specialization	4–6 Months
■ Cloud Security Basics	AWS/Azure/GCP shared responsibility model	
■ IAM & Zero Trust	Least privilege, MFA, conditional access	
■ Container Security	Docker, Kubernetes security best practices	
■ DevSecOps	SAST/DAST, CI/CD pipeline security	
■ Compliance & GRC	SOC 2, HIPAA, PCI-DSS, risk frameworks	
Recommended Certs: ■ AWS Security Specialty ■ CCSP ■ CISSP (advanced)		

ESSENTIAL TOOLS BY CATEGORY

Network Analysis	Wireshark · Nmap · tcpdump · Zeek · Netflow
SIEM & Monitoring	Splunk · Microsoft Sentinel · Elastic SIEM · IBM QRadar
Vulnerability Scan	Nessus · OpenVAS · Qualys · Rapid7 InsightVM
Incident Response	Autopsy · Volatility · FTK Imager · TheHive
Penetration Testing	Metasploit · Burp Suite · BloodHound · Cobalt Strike
Threat Intel	MISP · OpenCTI · VirusTotal · Shodan · AlienVault OTX

CAREER PATHS & SALARY RANGES (USD)

Role	Salary	Focus
SOC Analyst (Tier 1–3)	\$55K – \$95K	Monitor alerts, triage incidents, escalate threats
Incident Responder / DFIR	\$80K – \$130K	Lead IR investigations, forensics, malware analysis
Threat Intelligence Analyst	\$85K – \$125K	Produce actionable intel, track APTs, feed SIEM
Penetration Tester / Red Team	\$90K – \$150K	Authorized attacks, identify weaknesses, write reports
Cloud Security Engineer	\$110K – \$170K	Secure cloud infra, IAM policies, DevSecOps pipelines
Security Architect / CISO	\$140K – \$220K	Design enterprise security strategy & governance

CERTIFICATION ROADMAP

Level	Certification	Provider	Focus Area
Entry	CompTIA A+	CompTIA	IT fundamentals
Entry	CompTIA Network+	CompTIA	Networking
Entry	Google IT Support	Google/Coursera	IT support fundamentals
Intermediate	CompTIA Security+	CompTIA	Core security
Intermediate	CompTIA CySA+	CompTIA	SOC / threat analysis
Intermediate	eJPT	eLearnSecurity	Entry-level pen testing
Advanced	OSCP	Offensive Sec	Penetration testing
Advanced	GCIH	SANS/GIAC	Incident handling
Advanced	CCSP	(ISC) ²	Cloud security
Expert	CISSP	(ISC) ²	Security management
Expert	CISM	ISACA	Security management

PRO TIPS FOR SUCCESS

■ Build a Home Lab	Set up VMs with Kali Linux, Metasploitable, and a SIEM to practice daily.
--------------------	---

■ Practice on CTF Platforms	HackTheBox, TryHackMe, and PicoCTF build real skills faster than courses alone.
■ Document Everything	Maintain a personal wiki or GitHub repo with notes, writeups, and scripts.
■ Join the Community	Engage on r/netsec, DEFCON Discord, and local BSides events for networking.
■ Track Your Progress	Set 30/60/90-day goals. Review skills gaps quarterly and adjust your roadmap.
■ Stay Current	Follow threat intel feeds, read vendor advisories, subscribe to CISA alerts.

Remember: Cybersecurity is a marathon, not a sprint. Focus on deeply understanding concepts rather than collecting certifications. Hands-on experience and consistent practice are the true differentiators in this field. Stay curious, stay ethical, and keep learning.